

Business Trip Work Summary Report

Project Name: Analysis and Regulation of Cognitive Neural Circuits

Project No.: NSF-COG-2024-07 **Report Author:** Zhang Mingyuan

Report Date: July 25, 2024

I. Basic Travel Information

The undersigned traveled to the San Diego area, California, United States, from July 18 to 22, 2024, for a total of 5 days, to attend the 12th International Neuroscience Symposium. The accommodation period was from July 19 to July 21, 2024, totaling 2 nights.

II. Completion of Work Tasks

1. Academic Presentation: On the afternoon of July 20, delivered an invited presentation at the "Dynamic Imaging of Neural Circuits" sub-forum, titled "Working Memory Encoding Mechanisms of Prefrontal Cortex Based on Two-Photon Imaging." The presentation was well received by attending experts and generated multiple constructive feedback comments.
2. Collaboration Progress: On the evening of July 20, held a joint project progress meeting with Dr. Smith's team from Stanford University and Dr. Johnson's team from the Massachusetts Institute of Technology. Consensus was reached on the "Cross-Species Neural Circuit Comparative Study" experimental protocol, and meeting minutes were produced.

3. 3. Technical Research: On the morning of July 21, attended the "Pre-Clinical Evaluation of Brain-Computer Interfaces" technical workshop and conducted technical consultations with three equipment manufacturers, obtaining preliminary quotations and technical specifications for a new generation of multi-channel electrode arrays.
4. 4. Academic Exchange: During the conference, engaged in in-depth exchanges with 23 scholars from 12 countries and established new academic connections.

III. Outcomes and Follow-up Plans

1. 1. Produced one copy of "Meeting Minutes with Stanford/MIT Collaboration Team."
2. 2. Obtained three sets of equipment technical materials, providing a basis for the upgrade of the project's experimental platform.
3. 3. Based on conference exchange outcomes, initiated the revision of the project's second-half experimental plan.
4. 4. Reached preliminary intentions with two overseas young scholars for joint postdoctoral training.

IV. Expense Statement

During this trip, accommodation was arranged at a hotel near the conference venue. The accommodation period was from July 19 to July 21, 2024, totaling 2 nights. The accommodation expenses and related

taxes were paid by credit card. Please refer to the attached original invoices for details.

Report Author Signature: Zhang Mingyuan **Date:** July 25, 2024